DATASCI 406 FA 26 — Module 2 Practice Exam

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This practice exam asks about the Counting Customers practice project. Exam 2 has the same format and asks the same kinds of question about your own project, so work this one with your printed practice project in front of you.

80 minutes. 50 points. Paper notes only. No electronic devices.

Part A: Inversion (8 points)

Question 1 (1 point)

`rquit` takes the ceiling of $\log U / \log(1-p)$. Without the ceiling, what would it draw? Circle one.

1. An exponential with rate $-\log(1-p)$.
2. The same geometric distribution.
3. The geometric distribution shifted down by one.
4. Negative values, since $\log U < 0$.

Solution

Answer: 1. $-\log U$ is Exp(1), and dividing by $-\log(1-p) > 0$ makes it exponential with that rate. Option 4 forgets that $\log(1-p)$ is negative too.

Question 2 (1 point)

In Part A your QQ plot for J has many points stacked at the same integer heights, not a smooth line. Does this mean the sampler is wrong? Circle one.

1. No. Both sets of quantiles are integers, so many points coincide. The check is whether they sit on the 45 degree line.
2. Yes. A correct sampler gives a smooth line.
3. Yes. Discrete variables cannot be checked with a QQ plot.
4. No, but only because 10,000 draws is a large sample.

Solution

Answer: 1. A QQ plot for a discrete variable is a set of steps. The number of draws does not change that.

Question 3 (6 points)

(i) (2 points) Show that if E is exponential with rate $-\log(1-p)$, then $\lceil E \rceil$ has the distribution of J. Hint: find $P(\lceil E \rceil > j)$.

(ii) (2 points) Suppose in Part A you had compared your draws with `qgeom(u, p)`, without the +1. Describe what the QQ plot would show, and what you would conclude from it.

(iii) (2 points) Part A checked your sampler against `qgeom`, which R provides. Describe two ways to check a sampler for a distribution R has no functions for at all.

Solution

(i) 2 pts

- 1 pt: for an integer j, $\lceil E \rceil > j$ exactly when $E > j$.
- 1 pt: $P(E > j) = e^{j \log(1-p)} = (1-p)^j = P(J > j)$. Equal tails for every j mean equal distributions.

(ii) 2 pts

- 1 pt: the points lie on a line parallel to the 45 degree line, one unit above it.
- 1 pt: the right shape, shifted by one. The sampler and the reference count different things (purchases versus failures before quitting), not a broken sampler.

(iii) 2 pts, 1 for each. Examples: a QQ plot against a quantile function you derived yourself; plotting $F(X)$ for the draws against $U(0,1)$ (the probability integral transform); comparing the fraction of draws in a few intervals with the exact probabilities; for a discrete variable, comparing a table of frequencies with the PMF.

Part B: Mixtures (8 points)

Question 4 (1 point)

In Part B you drew p and then $J \mid p$. What kind of construction is the distribution of J? Circle one.

1. A mixture, with a random parameter p.
2. A sum of independent geometric variables.
3. A convolution of a beta and a geometric, $p + J$.
4. A transformation $J = h(p)$ for a fixed function h.

Solution

Answer: 1. Option 4 fails because the same p gives different values of J.

Question 5 (1 point)

In Part B the mean of your 10,000 draws of J was about 195, and the largest was over 330,000. You rerun with 1,000,000 draws. What happens to the sample mean? Circle one.

1. It does not settle down. It tends to be larger, and depends on the few largest draws.
2. It settles near 195.
3. It settles near $1/\mathrm{E}(p) = (a+b)/a \approx 4$.
4. It settles near $\mathrm{E}(1/p)$, a finite number.

Solution

Answer: 1. $\mathrm{E}(J)$ is infinite because $a < 1$, so the law of large numbers has nothing to converge to.

Question 6 (6 points)

(i) (2 points) If every customer had the average quitting probability $a/(a+b) \approx 0.25$, the chance of staying past 100 purchases would be $0.75^{100} \approx 5 \times 10^{-13}$. Part B found $P(J > 100) \approx 0.05$. Explain why letting p vary across customers makes long stays so much more likely.

(ii) (2 points) Given p, $\mathrm{E}(J \mid p) = 1/p$. Show that $\mathrm{E}(J) = \mathrm{E}(1/p)$ is infinite when $p \sim \mathrm{Beta}(a, b)$ with $a < 1$. Hint: look at the integral near $p = 0$.

(iii) (2 points) Part B checked the tail $P(J > j)$ at $j = 1$, 10 and 100 against the formula, and did not compare the sample mean with anything. Why is the mean useless as a check here, and why do the tail probabilities work?

Solution

(i) 2 pts

- 1 pt: for a customer with quitting probability p the chance of staying past 100 purchases is $(1-p)^{100}$, which is near 0 unless p is near 0.
- 1 pt: with $a < 1$ the Beta puts many customers at p near 0, and those customers stay a long time. Averaging $(1-p)^{100}$ over customers gives 0.05, not $(1-\bar{p})^{100}$.

(ii) 2 pts

- 1 pt: $\mathrm{E}(1/p) = \frac{1}{B(a,b)} \int_0^1 p^{a-2}(1-p)^{b-1}\, dp$.
- 1 pt: near 0 the integrand behaves like p^{a-2}, and $a - 2 < -1$, so the integral diverges.

(iii) 2 pts

- 1 pt: the mean is infinite, so the sample mean does not converge and there is no number to compare it with.
- 1 pt: each tail probability is the mean of a 0/1 indicator, which has finite variance and a binomial confidence interval. The simulated tails matched the formula: 0.759, 0.270 and 0.050 against 0.754, 0.267 and 0.050.

Part C: Accept-reject (8 points)

Question 7 (1 point)

In Part C a candidate Y is accepted when $U \leq (1-Y)^{b-1}$. Which candidates are most likely to be rejected? Circle one.

1. Candidates near 1.
2. Candidates near 0.
3. Candidates near the mode of f.
4. None. Every candidate has the same chance.

Solution

Answer: 1. With $b > 1$, $(1-Y)^{b-1}$ is near 0 when Y is near 1. The candidate density ap^{a-1} puts more weight near 1 than f does, and accept-reject removes the excess.

Question 8 (1 point)

Could importance resampling with the same candidate density $g(p) = ap^{a-1}$ have drawn from the Beta in Part C? Circle one.

1. Yes. g is positive wherever f is, and the weights f/g are bounded.
2. No. Importance resampling needs uniform candidates.
3. No. Importance resampling needs the normalizing constant $B(a, b)$.
4. No. f is unbounded near 0.

Solution

Answer: 1. The weight is proportional to $(1-p)^{b-1}$, at most 1. The constant cancels when the weights are normalized, and f being unbounded is the problem g was chosen to solve.

Question 9 (6 points)

(i) (2 points) In Part C you accepted 4,759 of 10,000 candidates, with a 95% interval of $(0.466, 0.486)$, against $1/c = aB(a, b) \approx 0.475$. Is the difference between 0.476 and 0.475 a sign of a problem? What would a problem look like?

(ii) (2 points) Suppose b had been 0.5 instead of 2.426, with a unchanged. Does the bound on f/g in Part C still hold? What goes wrong, and what kind of candidate density might work instead?

(iii) (2 points) Part C drew the Beta by accept-reject but drew the candidate g by inversion. Why is inversion the natural choice for g but not for the Beta?

Solution

(i) 2 pts

- 1 pt: no. The interval contains 0.475.
- 1 pt: a problem would be an interval that clearly misses $1/c$, which points to a wrong c or a wrong acceptance test, or accepted draws whose QQ plot against `qbeta` leaves the line.

(ii) 2 pts

- 1 pt: no. With $b < 1$, $(1-p)^{b-1}$ goes to infinity as $p \to 1$, so f/g has no bound and no c exists.
- 1 pt: f now goes to infinity at both ends, so the candidate must too. One choice is a 50/50 mixture of ap^{a-1} and $b(1-p)^{b-1}$, which can be drawn by picking a component and inverting. Any candidate that is unbounded at both ends earns the point.

(iii) 2 pts

- 1 pt: g has CDF p^a, which inverts in closed form to $u^{1/a}$.
- 1 pt: the Beta CDF is the incomplete beta function, which has no closed form, and neither does its inverse.

Part D: The purchase process (8 points)

Question 10 (1 point)

`purchases` draws each purchase time from the one before it. Which factorization of the joint distribution of the purchase times does this use? Circle one.

1. $f(t_1)f(t_2 \mid t_1)f(t_3 \mid t_2)\cdots$, each time depending only on the one before.
2. $f(t_1)f(t_2)f(t_3)\cdots$, since the times are independent.
3. $f(t_1 \mid t_2, t_3, \ldots)f(t_2 \mid t_3, \ldots)\cdots$, starting from the last purchase.
4. None. The times are drawn together as a vector.

Solution

Answer: 1. The gaps are independent; the times are not. Option 3 is a valid factorization, but not the one `purchases` uses.

Question 11 (1 point)

Your simulated data set in Part D had 1,386 customers with no repeat purchase, and the real data has 1,411. What would tell you whether a gap this size is a problem? Circle one.

1. Repeat the simulation many times and see whether 1,411 falls in the middle 95% of the simulated counts.
2. A t-test comparing the simulated and real counts of repeat purchases.
3. One simulation with ten times as many customers.
4. Nothing. Any gap means the model is wrong.

Solution

Answer: 1. This is a Monte Carlo test, as in module 1. Option 3 answers a different question, since the real data has 2,357 customers.

Question 12 (6 points)

(i) (2 points) Over t weeks an active customer's number of purchases is Poisson(λt), which R can draw in one call. Why does `purchases` draw the gaps one at a time instead?

(ii) (2 points) Suppose you repeated the Part D simulation 100 times and took the middle 95% of the number of customers with no repeat purchase. This treats the paper's r, α, a and b as the truth, but they are estimates from these same 2,357 customers. Would the range be wider or narrower if it allowed for that? Explain.

(iii) (2 points) Your Part D table matched the real counts of repeat purchases closely. Name one thing the model claims about customers that this bar chart does not check, and say where in the project it is checked.

Solution

(i) 2 pts

- 1 pt: the customer may quit after any purchase, so the purchases have to be generated in order, with a quitting decision after each one.
- 1 pt: a single Poisson count ignores quitting. It would also not give t_x, the time of the last purchase, which Part E needs.

(ii) 2 pts

- 1 pt: wider.
- 1 pt: other plausible values of the parameters would shift the count up or down, which adds to the variation from customer randomness alone.

(iii) 2 pts

- 1 pt: any claim beyond the count of repeat purchases. Examples: when the last purchase happened (t_x); how many purchases customers make after the calibration period; how the count and the timing go together.
- 1 pt: the prediction at the end of the project checks holdout purchases for customers with $x = 2$. Credit "not checked anywhere" for a claim that truly is not.

Part E: Gibbs sampling (8 points)

Question 13 (1 point)

The posterior of (λ, p) for one customer is not a standard distribution. Why could Part E still use a Gibbs sampler? Circle one.

1. With Z added, each full conditional is a standard distribution.
2. The posterior is a gamma times a beta.
3. Gibbs sampling does not need the full conditionals.
4. Customer 1 has only two repeat purchases.

Solution

Answer: 1. Adding a latent variable so that every conditional is easy is the reason Z is there. Option 2 describes the prior.

Question 14 (1 point)

Part E checked the Gibbs sampler against a closed form. Without one, you could check it by importance resampling: draw (λ, p) from the priors and weight each draw by the likelihood. For which customer would this work worst? Circle one.

1. A customer with many repeat purchases, like customer 6 with $x = 7$.
2. A customer with no repeat purchases.
3. Customer 1.
4. It works equally well for all of them.

Solution

Answer: 1. The prior puts most customers at low rates. A customer with seven purchases has a likelihood concentrated far from where most prior draws fall, so a few candidates carry nearly all the weight, as for the Challenger posterior in video 5. A customer with no repeat purchases has a likelihood that favors low rates, which is where the prior already puts most of its draws.

Question 15 (6 points)

(i) (2 points) Explain in words why

$$P(Z = 1 \mid \lambda, p) = \frac{(1-p)\, e^{-\lambda(T-t_x)}}{p + (1-p)\, e^{-\lambda(T-t_x)}}$$

gets smaller as $T - t_x$, the time since the last purchase, gets longer. What do the two terms in the denominator stand for?

(ii) (2 points) At the end of the project, the model predicted about 1.04 holdout purchases per customer with $x = 2$, and they made 1.39. The paper's equation (10), computed with no simulation at all, also gives 1.04. Is the gap Monte Carlo error, a mistake in the simulation code, chance in which 214 customers we happened to observe, or the model? Use the fact that the standard deviation of `x_star` among these customers is 1.84 ($\sqrt{214} \approx 14.6$).

(iii) (2 points) Describe one way the model could be wrong that would make customers with $x = 2$ buy more in the holdout period than it predicts.

Solution

(i) 2 pts

- 1 pt: the two terms are the two ways to see no purchase after t_x: the customer quit after the last purchase (probability p), or stayed and by chance made no purchase for $T - t_x$ weeks (probability $(1-p)\,e^{-\lambda(T-t_x)}$).
- 1 pt: the second term shrinks as the silence gets longer, so quitting becomes the better explanation.

(ii) 2 pts

- 1 pt: not Monte Carlo error or a coding mistake, because equation (10) agrees with the simulation without using it.
- 1 pt: the standard error of the actual mean is about $1.84/14.6 \approx 0.13$, and the gap of 0.35 is almost three of those, so chance is an unlikely explanation. That leaves the model.

(iii) 2 pts for a change with a mechanism that raises holdout purchases. Examples: customers who quit can come back; purchase rates rise over time or in the holdout season; customers who buy faster are also less likely to quit, so λ and p are dependent.

Part F: Big picture (10 points)

Question 16 (5 points)

Choose a sampler. For each target below, name the method from this module you would try first, and say in a phrase why. One point each.

1. The Pareto distribution, with $F(x) = 1 - x^{-k}$ for $x \geq 1$.
2. The number of purchases a customer makes in 39 weeks under the full model, given `rgamma`, `rbeta` and your `purchases`.
3. A point chosen uniformly in the unit disk, given `runif`.
4. A pair $(\log \lambda, \operatorname{logit} p)$ that is bivariate normal with correlation 0.3, given `rnorm`.
5. The posterior of (r, α, a, b) given all 2,357 customers, known only up to a constant, with no full conditional you can draw from.

Solution

1 pt each for a method that works with a reason that fits it.

1. Inversion: $Q(u) = (1-u)^{-1/k}$.
2. A mixture, with the customer's λ and p as random parameters. Draw them, then simulate the purchases.
3. Rejection: draw points in the square $[-1, 1]^2$ and keep those inside the disk, about 79% of them.
4. Cholesky: two independent normals times C, plus the mean.
5. Metropolis-Hastings: it needs only the unnormalized posterior.

Question 17 (5 points)

Find an example. Point to one place in the practice project that does each of the following, and say in a phrase what makes it one. One point each.

1. A sampler for a discrete distribution by inversion.
2. A latent variable that is never observed.
3. A sampler that rejects some of its candidates.
4. A process simulated one step at a time, each step given the one before.
5. Checking a Monte Carlo answer against something computed another way.

Solution

1 pt each, generously.

1. Part A, `rquit`, from the quantile function of J.
2. Z in Part E, whether the customer is still active; λ and p in Parts B and D, which are never observed for any customer.
3. Part C, `rbeta_ar`.
4. Part D, `purchases`; the Gibbs chain in Part E.
5. Part A against `qgeom`; Part B against the tail formula; Part C against `qbeta` and $1/c$; Part D against the real counts; Part E against the closed-form probability; the holdout prediction against equation (10).